

Dong Heon Han

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Robotics Department, University of Michigan, Ann Arbor, MI 48109, USA

SUMMARY

A PhD student developing large-scale electromagnetic platforms for microrobot swarms, focusing on collective control, swarmalator dynamics, and programmable morphing

EDUCATION

- **University of Michigan** 2029
PhD in Robotics Ann Arbor, MI
- **University of Michigan** 2025
MS in Mechanical Engineering Ann Arbor, MI
- **Georgia Institute of Technology** 2021
BS in Mechanical Engineering Atlanta, GA

WORK EXPERIENCE

- **University of Michigan | Robotics Department** June 2025 - Present
Research Assistant | PI: Dr. Steven Ceron Ann Arbor, MI
- **University of Michigan | Mechanical Engineering Department** Aug 2023 - May 2025
Research Assistant | PI: Dr. Daniel Bruder Ann Arbor, MI
- **Republic of Korea Army** Jan 2022 - July 2023
Signal Specialist Korea
- **Seoul National University | Biosystems Engineering Department** Aug 2021 - Dec 2021
Research Assistant | PI: Dr. Young-Jun Park Seoul, Korea
- **George W. Woodruff School of Mechanical Engineering** Aug 2019 - Aug 2021
Research Assistant | PI: Dr. Ye Zhao Atlanta, GA
- **Korean Institute of Machinery and Materials** Jun 2018 - Aug 2018
Research Intern Daejeon, Korea

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, PAT=PATENT, S=IN SUBMISSION, P=PREPARATION, T=THESIS

- [C.1] **D.H. Han**, X. Xu, Y. Chen, Y. Zhou, X. Zhang, D. Bruder, X. Huang. "Towards Whole-Body Proprioceptive Morphing: A Soft Gripper for Cross-Scale Grasping". 2026 IEEE International Conference on Soft Robotics (RoboSoft).
- [C.2] R. Zuo, M. Mehta, **D.H. Han**, D. Bruder. "Embedded Valves for Distributed Control of Soft Pneumatic Actuators". 2024 IEEE International Conference on Intelligent Robots and Systems (IROS)
- [C.3] **D.H. Han**, S.J. Byeon, K.D. Kim, G.H. Han, M.H. Cha, Y.J. Park. "Development of Path Tracking Control Algorithm for Tractor Autonomous Driving". 2021 Korean Society for Agricultural Machinery Conference
- [Pat.1] **Blowers With Variable Nozzles**. US 11668311 B2. Issued June 6, 2023.
- [S.1] **D.H. Han**, M. Mehta, R. Zuo, Z. Wanger, and D. Bruder. "An Enhanced Proprioceptive Method for Soft Robots Integrating Bend Sensor and IMU"
- [S.2] R. Zuo, **D.H. Han**, R Li, S. Jamal, D. Bruder. "UMArm: Untethered, Modular, Wearable, Soft Pneumatic Arm with 12 DoF"
- [P.1] **D.H. Han**, R. Zuo, C. Kim, S. Jamal, D. Bruder. "BendSense: A Modular Off-the-Shelf Bend Sensor Framework for Proprioception in Soft Robots"

HONORS AND AWARDS

- **VIP Innovation Competition, 1st Place in Hardware, Devices & Robotics Track** Apr 2021
Georgia Institute of Technology
◦ Awarded to the most innovative and active research team in Georgia Tech
- **President's Undergraduate Research Award** Oct 2020
Georgia Institute of Technology
◦ Research excellence scholarship as an undergraduate researcher at Georgia Tech
- **Georgia Korean American Grocers Association Scholarship Award** Dec 2016
KAGRO
◦ Awarded for academic excellence and leadership in community service
- **Kappa Mu Epsilon** May 2018
Mathematics honor society

CORE SKILLS

- **Robotics:** System-level robotic design and integration, hardware/firmware/software co-design, ROS2
- **Programming:** C, C++, Python, MATLAB
- **Design and Fabrication:** CAD, 3D printing, silicone casting, rapid prototyping

SERVICE

- **Reviewer**
 - IEEE Transactions on Mechatronics
 - IEEE Robotics and Automation Letters
 - IEEE International Conference on Robotics and Automation (ICRA)
 - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
 - IEEE International Conference on Soft Robotics (RoboSoft)
- **Military Service**
 - Republic of Korea Army, Signal Specialist Sergeant
- **Books for Africa**
 - Led the initiative to carefully select, categorize, and distribute educational books to underprivileged regions in Africa, contributing to the advancement of literacy and learning opportunities

TEACHING EXPERIENCE

- **Heat Transfer Tutor** *May 2021 - Aug 2021*
Georgia Institute of Technology
 - Taught Georgia Tech undergrad in a tutoring program sponsored by Shell Oil and Air Products

REFERENCES

1. **Dr. Steven Ceron**
Assistant Professor, Robotics | *University of Michigan*
2. **Dr. Daniel Bruder**
Assistant Professor, Mechanical Engineering | *University of Michigan*
3. **Dr. Ye Zhao**
Associate Professor, Mechanical Engineering | *Georgia Institute of Technology*